

MD. RAIHAN ALI

Machine Learning & AI Engineer

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🌐 Raihan.A 🐙 Github 🌐 Portfolio63

👤 PROFILE

Computer Science Engineer specializing in Machine Learning, AI Automation, and Federated Learning. Proven ability to design and deploy end-to-end AI pipelines, automate workflows, and build practical solutions for real-world challenges.

📁 EXPERIENCE

Research Ustad 🏠, *Moderator & Team Lead* 05/2026 – Present

- Led research teams and contributed to ongoing studies as an active researcher.
- Reviewed member submissions to ensure alignment with platform standards and research integrity guidelines.
- Enforced community guidelines to maintain a professional, collaborative research environment.

Data Science Intern, *HexSoftwares* 10/2025 – 11/2025

- Collected, cleaned, and preprocessed real-world datasets for machine learning model development.
- Developed and fine-tuned ML models, evaluating performance against standard metrics to improve accuracy.

Machine Learning Intern, *CodeAlpha* 07/2025 – 09/2025

- Built and trained ML models to extract insights that supported data-driven business decisions.
- Performed exploratory data analysis and feature engineering to improve model performance.

🎓 EDUCATION

B.Sc in Computer Science and Engineering 2022 – 2026

University of Dhaka (Faculty of Engineering and Technology, NITER)

- CGPA: 3.60/4.00 Dhaka, Bangladesh

📖 PUBLICATIONS

[J1] Federated and Explainable AI for Population and Life Expectancy Prediction Accepted
IEEE Journal (Q1)

Proposes a federated learning framework combined with explainable AI techniques to predict population trends and life expectancy while preserving data privacy across distributed sources.

Conference

[C1] FuzzyTriage-BD: Explainable Fuzzy Triage for Bangla Telemedicine Specialist Routing Accepted

MD. Raihan Ali^{1st}, Md Sohag Hossain^{2nd}, Sagor Das^{3rd}, MD Tanjum An Tashrif^{4th}
IEEE BECITHCON 2026, IUBAT, Dhaka

Presents an explainable fuzzy-logic triage system with BanglaBERT-based named entity recognition to route Bangla telemedicine queries to the appropriate medical specialist.

[C2] Federated Explainable Learning for Privacy-Preserving Antenatal Care Quality Prediction in Bangladesh Accepted

Md Sohag Hossain^{1st}, MD. Raihan Ali^{2nd}, Sagor Das^{3rd}, MD Tanjum An Tashrif^{4th*}
IEEE BECITHCON 2026, IUBAT, Dhaka

Applies privacy-preserving federated learning with explainable AI to predict antenatal care quality in Bangladesh without centralizing sensitive patient data.

📁 PROJECTS

Time-Series Forecasting of Carbon Monoxide and Nitrogen Dioxide Levels

- Developed machine learning models to forecast daily and hourly concentrations of carbon monoxide and nitrogen dioxide.
- Engineered time-based features from historical air quality data to improve forecasting accuracy across multiple prediction horizons.

Sentiment Analysis on Amazon Product Reviews

- Built a natural language processing model to analyze customer sentiment from Amazon product reviews and classify feedback polarity.
- Applied text preprocessing and vectorization techniques to prepare unstructured review data for model training.

Handwritten Character Recognition

- Implemented a deep learning model using CNNs to recognize handwritten characters and digits with high accuracy.
- Trained and validated the model on image data, applying augmentation techniques to improve generalization.

COURSES & CERTIFICATIONS

- Data Science & ML

ID: 32c6f51

- Data Engineering

ID: a49c3d8784

- Data Visualization

ID: 7843-3754-9729

- Google Data Analytics

ID: 5V71917MZBDB

- Advanced Data Analytics

ID: LZLIoIDVD8LF

- Data for Exploration

ID: MII6V23HDNLF

- Intro to Data Science

IBM

- Robotic Camp 2021

ID: 22020224

SKILLS

Programming Languages

Python | C++ | SQL | Bash

AI Agent Frameworks & Workflows

LangChain | LangGraph | Agent-based Systems | Prompt Engineering | Retrieval-Augmented Generation (RAG)

LLM Integration

OpenAI GPT Models | LLaMA | Mistral | Open-Source & Cloud-Based LLMs

Machine Learning & Deep Learning

TensorFlow | PyTorch | Scikit-learn | CNN | RNN/LSTM | Transformers | Federated Learning

Workflow Automation

n8n | Zapier | Make

Data Analytics & Visualization

Pandas | Matplotlib | Seaborn | Power BI | Excel

Tools & Version Control

Git/GitHub | Jupyter | VS Code

Research & Soft Skills

Academic Research & Publication | Technical Writing | Problem Solving | Adaptability to Fast-Paced Environments | Team Collaboration

LEADERSHIP & COMMUNITY INVOLVEMENT

Volunteer, International General Science Olympiad (IGSO)

2025

- Promoted the Science Olympiad through live sessions at top schools and colleges in Savar.

Captain, Binary Football League, Dept. of CSE, NITER

2025

- Led and captained a team representing the Department of Computer Science and Engineering in an intra-department football league, coordinating team strategy and player management.

REFERENCE

Anichur Rahman

Assistant Professor, Dept. of CSE

National Institute of Textile Engineering & Research (NITER)

Georgia Southern University, USA

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